

Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiment 4

Comparative Study of Deep Convolutional Neural Network Architectures Using Transfer Learning

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

1. Objective

The objectives of this experiment are

- Study the evolution of deep CNN architectures.
- Compare LeNet-5, AlexNet, VGG16, GoogleNet and ResNet.
- Understand transfer learning.
- Fine tune pretrained CNN models.
- Compare classification performance of different architectures.

2. Learning Outcomes

After completing this experiment students will be able to

1. Explain modern CNN architectures.
2. Differentiate LeNet, AlexNet, VGG16, GoogleNet and ResNet.
3. Implement transfer learning.
4. Fine tune pretrained CNN models.
5. Compare CNN architectures using performance metrics.

3. Introduction

Deep Convolutional Neural Networks have become the dominant models for computer vision applications because they automatically learn hierarchical features from images. The evolution of CNNs has focused on improving classification accuracy while reducing computational complexity and training difficulties.

The progression of CNN architectures began with LeNet-5 and has evolved through AlexNet, VGGNet, GoogleNet, and ResNet. Each architecture introduced important innovations that significantly improved image classification performance.

4. Evolution of CNN Architectures

Model	Year	Major Contribution
LeNet-5	1998	First practical convolutional neural network
AlexNet	2012	ReLU activation, Dropout and GPU training
VGG16	2014	Uniform 3×3 convolution filters
GoogleNet	2014	Inception modules for multi-scale feature extraction
ResNet	2015	Residual learning using skip connections

5. LeNet-5

LeNet-5 was proposed by Yann LeCun in 1998 for handwritten digit recognition. It consists of two convolution layers, two average pooling layers and fully connected layers. It demonstrated that convolutional networks could automatically learn image features without handcrafted feature extraction.

Advantages

- Small number of parameters
- Fast training
- Suitable for OCR

6. AlexNet

AlexNet won the ImageNet Challenge in 2012 by reducing the classification error significantly. It introduced ReLU activation, Dropout regularization and GPU training.

Major innovations

- ReLU activation
- Dropout
- GPU implementation
- Data augmentation

7. VGG16

VGG16 demonstrated that using multiple small 3×3 convolution filters could outperform larger filters while increasing network depth.

Advantages

- Excellent feature extraction
- Simple architecture
- High classification accuracy

8. Comparison of Architectures

Architecture	Depth	Parameters	Main Contribution
LeNet-5	7	60K	First CNN
AlexNet	8	61M	ReLU + Dropout
VGG16	16	138M	Deep 3×3 filters
GoogLeNet	22	6.8M	Inception Modules
ResNet50	50	25.6M	Residual Learning

9. GoogLeNet (Inception Network)

GoogLeNet, proposed by Szegedy *et al.* in 2014, introduced the **Inception Module**, which performs multiple convolution operations in parallel using different filter sizes. Instead of using only a single kernel size, GoogLeNet simultaneously applies 1×1 , 3×3 , 5×5 convolutions and max pooling. The outputs are concatenated to obtain multi-scale feature representations.

The use of 1×1 convolutions before larger kernels reduces the number of trainable parameters and computational complexity.

Advantages

- Multi-scale feature extraction.
- Fewer parameters than VGG16.
- High computational efficiency.
- Better classification performance.

10. ResNet

Increasing CNN depth often causes the **vanishing gradient problem**, making optimization difficult. ResNet solves this issue using **skip (residual) connections**, allowing gradients to flow directly through the network.

Instead of learning

$$H(x)$$

ResNet learns

$$F(x) = H(x) - x$$

thus,

$$Output = F(x) + x$$

where

- $F(x)$ = Residual Mapping
- x = Identity Mapping

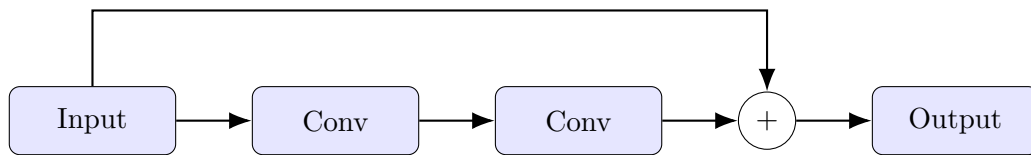


Figure 1: Residual Block in ResNet

Advantages

- Eliminates vanishing gradients.
- Enables very deep neural networks.
- Faster convergence.
- High ImageNet accuracy.

11. Dilated Convolution

Dilated convolution (also called atrous convolution) increases the receptive field without increasing the number of parameters.

The dilation rate determines the spacing between kernel elements.

For a standard convolution,

$$D = 1$$

For dilated convolution,

$$D > 1$$

Example

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & 0 \\ 4 & 0 & 5 \end{bmatrix}$$

where zeros indicate inserted gaps.

Applications

- Semantic segmentation
- Medical image analysis

- Satellite imagery
- Object localization

12. Transpose Convolution

Transpose convolution increases the spatial dimensions of feature maps. Unlike pooling, which reduces image size, transpose convolution performs learnable upsampling.

Applications

- Image Super Resolution
- Autoencoders
- GANs
- Image Generation
- Semantic Segmentation

13. Transfer Learning

Transfer Learning utilizes a pretrained CNN model that has already learned general image features from a large dataset such as ImageNet.

Instead of training from scratch,

1. Load a pretrained model.
2. Remove the original classifier.
3. Freeze convolution layers.
4. Add new Dense layers.
5. Train the classifier.
6. Fine tune selected convolution layers.

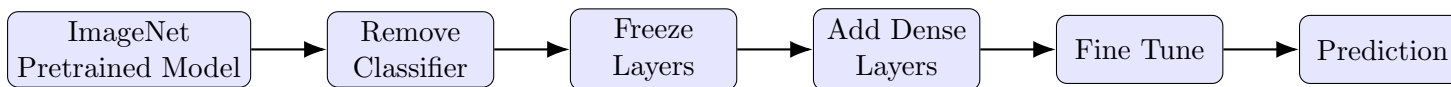


Figure 2: Transfer Learning Workflow

Advantages

- Faster convergence.
- Higher accuracy on small datasets.
- Reduced training time.
- Lower computational cost.

14. Dataset

Dataset: CIFAR-10

- Training Images : 50,000
- Testing Images : 10,000
- Number of Classes : 10
- Image Size : $32 \times 32 \times 3$
- Classes: Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship and Truck.

15. Experimental Procedure

Task 1: Dataset Preparation

1. Load the CIFAR-10 dataset using TensorFlow/Keras.
2. Normalize the pixel values to the range $[0,1]$.
3. Display ten sample images.

4. Print the dimensions of the training and testing datasets.

Task 2: Transfer Learning

Load any one of the following pretrained CNN models.

- VGG16
- ResNet50
- MobileNetV2
- EfficientNetB0 (Optional)

Perform the following steps.

1. Load pretrained ImageNet weights.
2. Remove the original classification layer.
3. Freeze the convolutional base.
4. Add a Global Average Pooling layer.
5. Add a Dense layer with ReLU activation.
6. Add the output layer with Softmax activation.

Task 3: Model Training

Train the model using

Optimizer	Adam
Learning Rate	0.001
Batch Size	32
Epochs	10–20
Loss Function	Categorical Cross Entropy
Evaluation Metric	Accuracy

Task 4: Fine Tuning

1. Unfreeze the last convolution block.
2. Train for another 5–10 epochs.
3. Compare the classification accuracy before and after fine tuning.

Task 5: Model Evaluation

Evaluate the trained model using

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- Classification Report

16. Hyperparameter Study

Compare the performance using different settings.

Hyperparameter	Values
Learning Rate	0.001, 0.0001
Batch Size	16, 32, 64
Epochs	10, 20
Optimizer	Adam, SGD
Dense Units	128, 256
Frozen Layers	All, Partial

17. Mandatory Plots

Include the following plots in the laboratory report.

1. Sample CIFAR-10 images
2. Training Accuracy
3. Validation Accuracy
4. Training Loss
5. Validation Loss
6. Confusion Matrix
7. Misclassified Images (Optional)



Figure 1: Sample Images

The plot shows sample images from the CIFAR-10 dataset belonging to different classes. The dataset contains RGB images of size 32*32 from 10 classes.

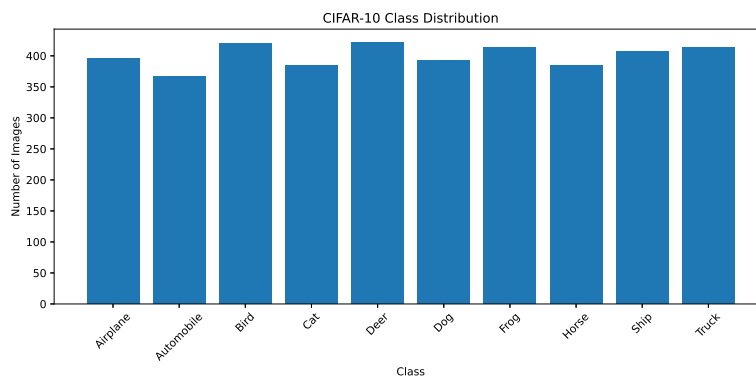


Figure 2: Class Distribution

The class distribution is approximately balanced across all 10 CIFAR-10 classes. Each class contains nearly the same number of images, so there is no major class imbalance.

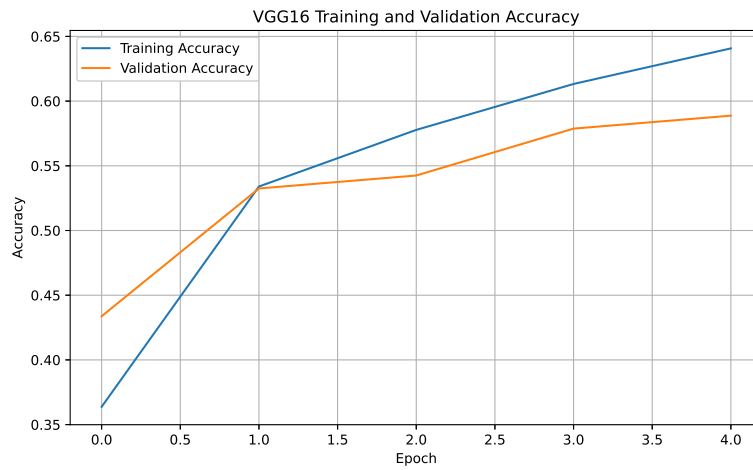


Figure 3: Training Accuracy

Training accuracy increases as the number of epochs increases, showing that the model learns the training data effectively. The high training accuracy compared with validation accuracy also indicates possible overfitting.

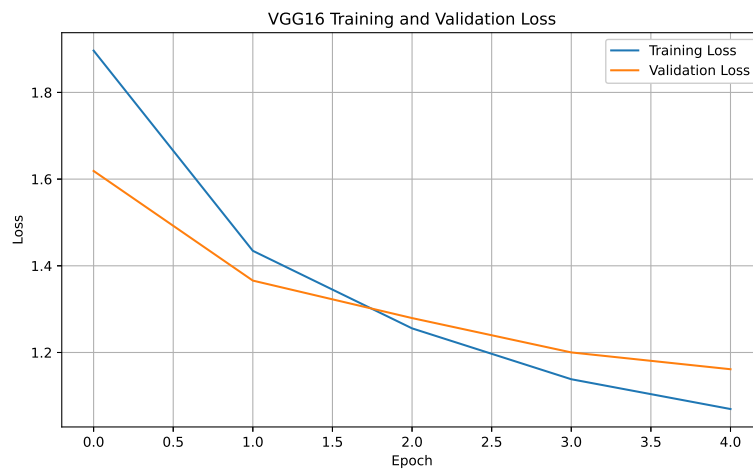


Figure 4: Training Loss

Training loss decreases as training progresses, indicating that the model is learning useful features from the images. The decreasing loss is consistent with the increase in training accuracy.

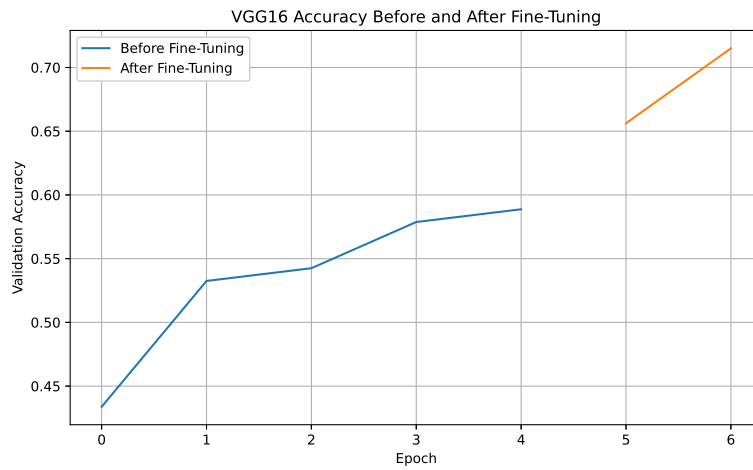


Figure 5: Fine Tuning Accuracy Comparison

Fine-tuning the pretrained model can improve its ability to adapt ImageNet-learned features to CIFAR-10. The improvement depends on the architecture and the amount of fine-tuning performed.

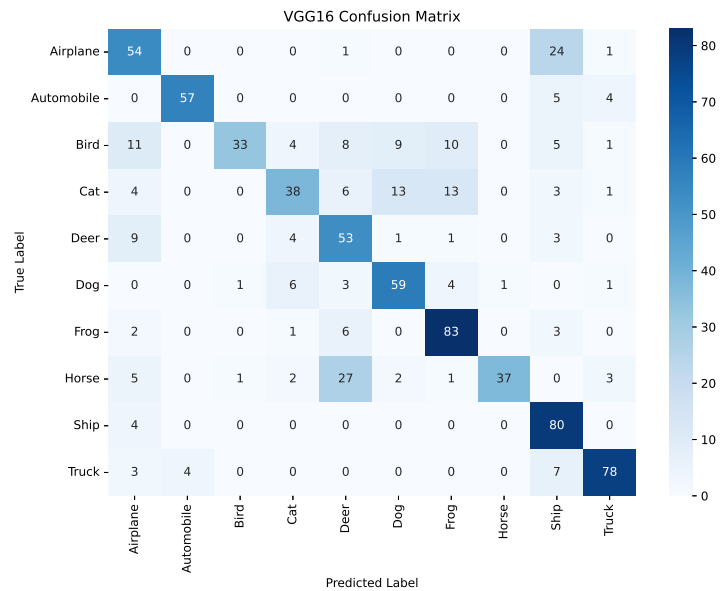


Figure 6: Confusion Matrix

The confusion matrix shows that some CIFAR-10 classes are classified more accurately than others. Classes with similar visual features have more misclassifications because they are harder for the model to distinguish.

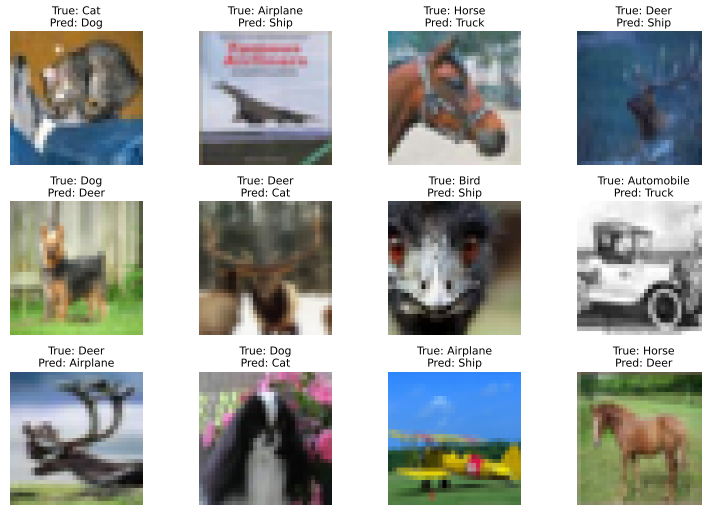


Figure 7: Misclassified Images

The misclassified images show cases where the model confuses visually similar classes. These errors indicate the difficulty of classifying low-resolution 32×32 images.

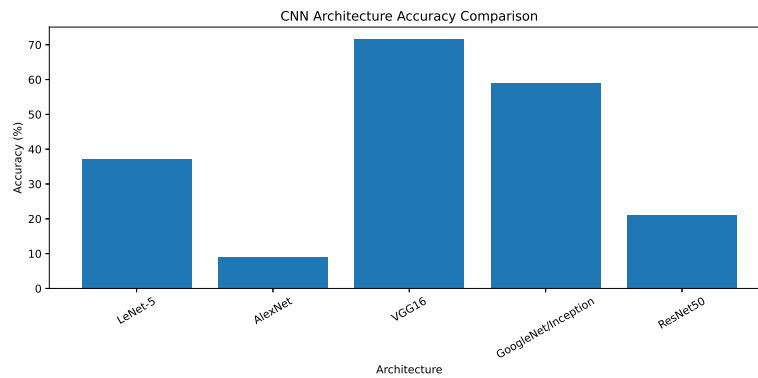


Figure 8: Architecture Accuracy Comparison

The comparison shows that different CNN architectures provide different classification performance and computational costs. Deeper architectures generally learn more complex features but require more parameters and training time.

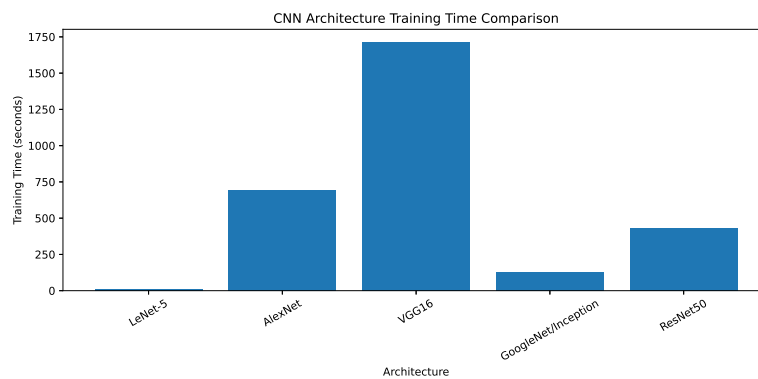


Figure 9: Architecture TrainingTime Comparison

More complex CNN architectures require greater training time because they contain more layers and parameters. This shows the trade-off between model complexity and computational efficiency.

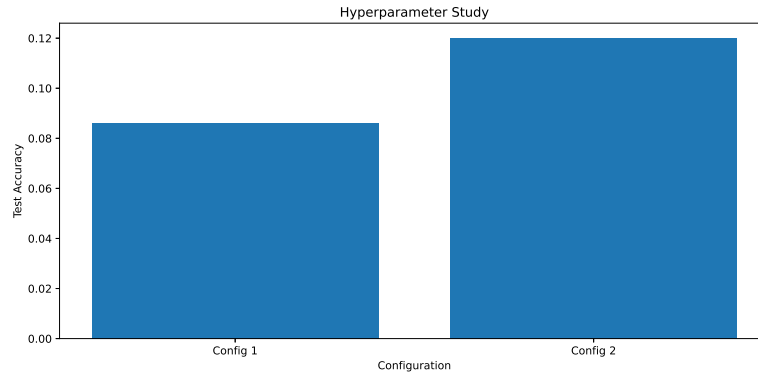


Figure 10: Hyperparameter Accuracy Plot

The plot shows that the model accuracy changes with different hyperparameter combinations. The best-performing combination gives the highest test accuracy, showing that suitable learning rate, batch size, optimizer, and dense units improve model performance.

18. Results

18.1 Performance Metrics

Metric	Value
Training Accuracy	0.64075
Testing Accuracy	0.71500
Precision	0.75803
Recall	0.71500
F1-score	0.70855
Training Time (s)	1716.22
Total Parameters	14,781,642

18.2 Comparison of CNN Architectures

Model	Parameters	Accuracy (%)	Training Time (s)
LeNet-5	62,006	37.125	9.49
AlexNet	36,925,322	8.875	694.65
VGG16	14,781,642	71.500	1716.22
GoogleNet	22,066,346	58.875	126.53
ResNet50	23,851,274	21.125	430.50

19. Discussion Questions

1. **Why is AlexNet considered a breakthrough in deep learning?**
AlexNet showed that deep CNNs with GPUs could achieve excellent image classification performance. It also popularized ReLU activation, dropout, and data augmentation in deep learning.
2. **Why does VGG16 use only 3×3 convolution filters?**
Small 3×3 filters allow VGG16 to learn complex features using multiple convolution layers. They also use fewer parameters compared with larger filters while maintaining a similar receptive field.
3. **Explain the advantages of the Inception module.**
The Inception module applies different convolution sizes and pooling operations in parallel. This allows the network to capture features at different scales while using computational resources efficiently.
4. **What is the purpose of residual learning?**
Residual learning helps very deep networks learn by using shortcut connections between layers. It reduces the vanishing-gradient problem and makes deep networks easier to train.
5. **Differentiate LeNet and ResNet.**
LeNet is a small CNN designed mainly for handwritten digit recognition. ResNet is a much deeper architecture that uses residual connections to enable effective training of deep networks.
6. **What is Transfer Learning?**
Transfer learning uses a model pretrained on a large dataset such as ImageNet for a new task. The learned features are reused, reducing the amount of training required for the new dataset.
7. **Why is fine tuning required?**
Fine tuning adapts the pretrained model's learned features to the specific target dataset. Un-freezing some deeper layers allows the model to learn features that are more suitable for the new task.
8. **Explain the difference between dilated convolution and transpose convolution.**
Dilated convolution expands the receptive field by inserting gaps between kernel elements without increasing the kernel size. Transpose convolution increases the spatial dimensions of feature maps and is commonly used for upsampling.
9. **Why do pretrained models converge faster?**
Pretrained models already contain useful features learned from a large dataset. Therefore, they start from a better initialization and require fewer updates to achieve good performance.
10. **Compare the computational complexity of LeNet and ResNet.**
LeNet has relatively few layers and parameters, so it requires much less computation and memory. ResNet is much deeper and has significantly higher computational requirements, although residual connections make its training effective.

20. Additional Exercises

1. Implement transfer learning using VGG16.
 - VGG16 transfer learning achieved a testing accuracy of 71.50%.
2. Repeat the experiment using ResNet50.
 - ResNet50 achieved an accuracy of 21.125% in the architecture comparison.
3. Compare Adam and SGD optimizers.
 - SGD achieved 12.00% test accuracy, higher than Adam's 8.625%.
4. Compare training with frozen layers and fine tuning.
 - Fine tuning improved validation accuracy from 58.875% to 71.50%.
5. Evaluate the model using Precision, Recall and F1-score.
 - The model achieved Precision 0.75803, Recall 0.71500 and F1-score 0.70855.
6. Compare the performance of LeNet, AlexNet and ResNet.
 - LeNet-5 achieved 37.125%, AlexNet 8.875% and ResNet50 21.125% accuracy.

21. References

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